Data Analyst Technical Assessment

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Contents

About This Report

This report presents the results of a technical assessment of a digital advertising campaign dataset. The objective of the analysis is to evaluate campaign performance, identify the factors associated with advertising effectiveness, and provide data-driven recommendations to improve Return on Advertising Spend (ROAS).

To accommodate both business stakeholders and technical reviewers, the report is organized into two complementary sections.

- **Main Report** — Presents the key business findings, interpretations, and recommendations in a concise, stakeholder-focused manner. Statistical results are summarized only where necessary to support business decisions.
- **Technical Appendices** — Provide the detailed analytical procedures supporting the main report, including the complete data quality assessment, data cleaning methodology, exploratory analyses, statistical assumption checks, hypothesis testing, and post-hoc analysis.

Readers interested primarily in the business implications may focus on the main report, while readers wishing to review the analytical methodology and supporting evidence may refer to the appendices.

This document also includes interactive cross-references. Figures, tables, equations, sections, and appendices references are hyperlinked throughout the report, allowing readers to navigate directly to the referenced content by clicking the corresponding reference.

Software Used

- Python 3.13.5 (data cleaning, statistical analysis, and data processing)
- Visual Studio Code (Python development environment)
- Microsoft Power BI (data visualization and dashboard development)
- TeXstudio (report preparation using LaTeX)

Task 1

Data Quality and Initial Exploratory Assessment

Before conducting the performance analysis, the advertising campaign dataset was assessed to ensure that it was suitable for business decision-making. A comprehensive data quality assessment was performed, including verification of data types, duplicate records, missing values, logical consistency, and potential outliers.

The complete details of these procedures are provided in Appendix A.

1.0.1 Data Cleaning Summary

The data cleaning process consist of:

- Verified variable data types
- Checked duplicate observations
- Removed observations(20 datapoints) with missing revenue.
- Retained structurally missing Video Completion Rate values.
- Retained valid view-through conversions.
- Removed observations(47 datapoints) with 0 clicks,impressions and purchases with non-zero purchase.
- Addressed severe `Spend` outliers.

Overall, the dataset was determined to be sufficiently complete and reliable for subsequent analysis.

1.0.2 Challenge

The primary business challenges include:

- Overall ROAS is substantially below the target value.
- Campaign performance varies considerably across advertising platforms, creative types, and product categories.

- Weekly ROAS has shown a declining trend, indicating decreasing advertising efficiency.
- Regional performance differences remain unclear.
- The impact of competitive events on campaign performance is uncertain.
- The effectiveness of different creative types and audience segments has not yet been established.
- Advertising spend must be reduced in underperforming areas while maintaining revenue growth.

1.0.3 Performance Metrics

We need numbers to evaluate campaign effectiveness. In that regard, four standard digital marketing performance metrics were considered.

Metric	Formula
CTR	Clicks / Impressions
CPC	Spend / Clicks
CVR	Purchases / Clicks
ROAS	Revenue / Spend

Table 1.1: Calculated performance metrics.

These metrics provide standardized measures of audience engagement, advertising efficiency, conversion effectiveness, and financial return.

1.1 Initial Exploratory Assessment

1.1.1 Business Overview

Following data cleaning, the overall Return on Advertising Spend (ROAS) is 0.96, which is below the company's target of 1.20. This indicates that the current advertising strategy is not generating sufficient revenue relative to advertising expenditure. Performance also varies across campaigns, platforms, creative types, product categories, and target audiences, suggesting opportunities to improve budget allocation and campaign effectiveness.

Since improving ROAS is the primary business objective, the subsequent analysis focuses on identifying the factors associated with both high- and low-performing advertising campaigns.

1.1.2 Key Performance Metrics

To evaluate advertising performance, the following baseline metrics were selected.

Metric	Business Relevance
Spend	Measures advertising investment and budget allocation.
Revenue	Measures the financial return generated by advertising activities.
ROAS	Primary indicator of advertising efficiency and campaign profitability.
CTR	Measures audience engagement with advertisements.
CPC	Measures the cost required to generate user engagement.
CVR	Measures the effectiveness of converting engaged users into customers.

Table 1.2: Baseline performance metrics used throughout the analysis.

Together, these metrics provide a comprehensive view of advertising performance, covering investment, engagement, conversion, and financial return.

1.1.3 Preliminary Insights

I created a power BI dashboard(file is inside github repository) to summarize the ROAS to gives us a main idea of the campaign ad ROAS across different categorical variables.

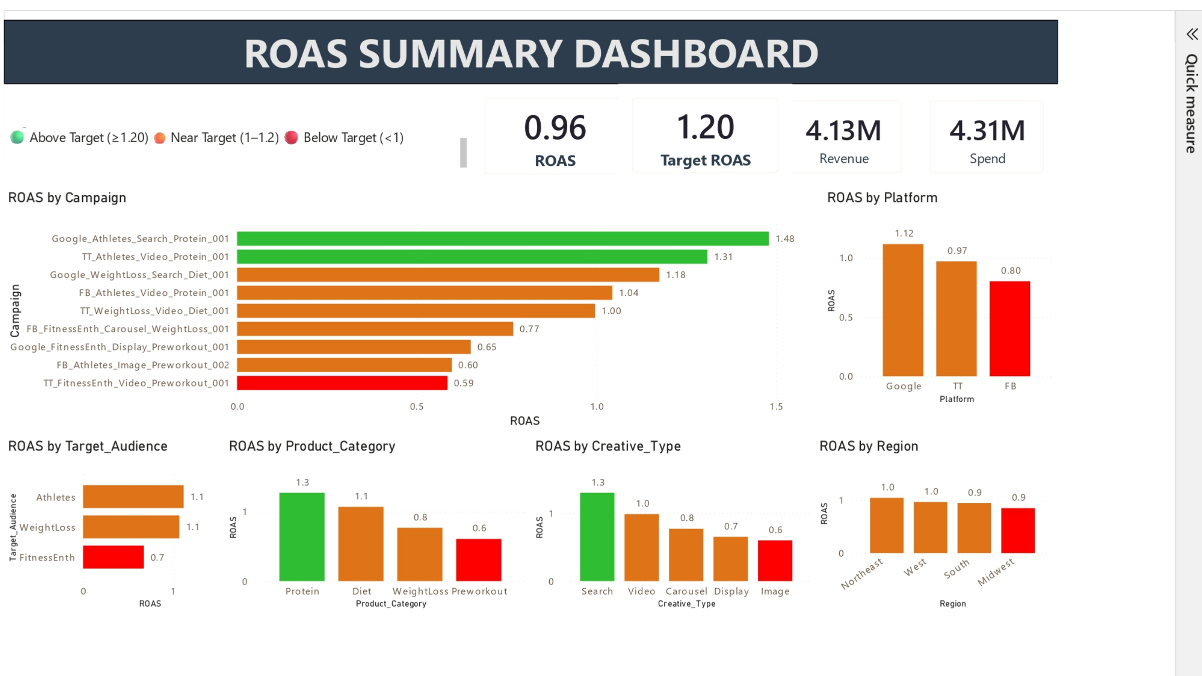


Figure 1.1: ROAS Dashboard Summary

The executive dashboard provides an initial overview of advertising performance prior to detailed analysis. Several observations can be made:

- Overall ROAS (0.96) remains below the business target of 1.20.

- Campaign performance varies substantially, with only a few campaigns exceeding the target ROAS.
- Google delivers the highest ROAS among advertising platforms, while Facebook records the lowest.
- Search creatives and Protein campaigns achieve the strongest advertising returns.
- Performance differences across regions are relatively small, whereas larger variation exists across creative types, product categories, and target audiences.

These observations establish a baseline understanding of current advertising performance and guide the subsequent analyses presented in Task 2.

Task 2

Performance Analysis

The exploratory analysis identified observable differences in performance metrics across multiple business dimensions, including advertising platform, region, creative type, product category, and target audience.

However, descriptive statistics alone cannot determine whether these observed differences represent genuine performance differences or are simply the result of random variation. Therefore, statistical significance testing was performed before drawing business conclusions.

2.1 Statistical Methodology

Preliminary assessment of the data showed that the distributions of several performance metrics, particularly ROAS, deviated significantly from normality. Such behavior is common in advertising datasets, where campaign performance is often positively skewed due to a small number of exceptionally high-performing campaigns.

Consequently, non-parametric statistical methods were adopted where appropriate.

The following statistical tests were used throughout the analysis:

1. **Shapiro–Wilk Test**

   Used to assess whether the distribution of each performance metric follows a normal distribution. The results determined whether parametric statistical methods were appropriate for subsequent analyses.

   Statistical Question:

   > *Do the observed performance metrics satisfy the assumptions required for parametric statistical tests?*

2. **Levene's Test**

   Used to determine whether the variability (consistency) of a performance metric differs significantly across groups.

   Business Question Answered:

   > *Which business groups deliver more consistent advertising performance, and which exhibit greater variability or unpredictability?*

3. **Kruskal–Wallis Test**

A non-parametric alternative to one-way ANOVA used to determine whether the distribution of a performance metric differs significantly across multiple groups. This test was selected because the normality assumption required for ANOVA was not satisfied.

Business Question Answered:

> *Do the observed differences in campaign performance across business groups (e.g., platforms, regions, creative types, product categories, and target audiences) represent genuine differences, or are they likely due to random variation?*

4. **Dunn's Post-hoc Test**

   Performed following a statistically significant Kruskal–Wallis test to identify which specific groups differ from one another while controlling for multiple comparisons.

   Business Question Answered:

   > *If significant differences exist, which specific business groups differ in their advertising performance?*

Together, these statistical procedures provide a systematic framework for validating assumptions, assessing performance consistency, determining whether differences are statistically significant, and identifying the specific groups responsible for those differences.

2.2 Channel Performance Analysis

The exploratory analysis revealed observable differences in advertising performance across Facebook, Google, and TikTok.

To determine whether these differences reflected genuine performance variation rather than random fluctuations, statistical significance testing was performed. A complete summary of the statistical analyses, including assumption checks, Kruskal–Wallis tests, and Dunn's post-hoc comparisons, is provided in Appendix B.1.

2.2.1 Return on Advertising Spend (ROAS)

Return on Advertising Spend (ROAS) was selected as the primary measure of advertising profitability. Among the three platforms, Google achieved the highest average ROAS (1.106), followed by TikTok (0.960), while Facebook recorded the lowest average ROAS (0.816) as shown in *Fig. 2.1*.

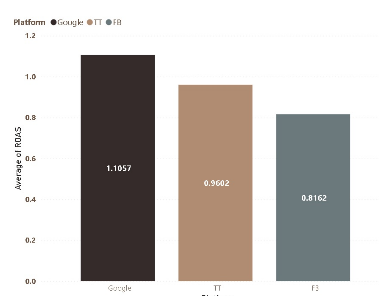


Figure 2.1: ROAS by Platform

Statistical testing confirmed that **these differences were significant** (Appendix B.1.1), indicating that Google consistently generated greater revenue for each dollar invested in advertising than the other platforms.

2.2.2 Click-Through Rate (CTR)

Click-Through Rate (CTR) was analyzed to evaluate each platform's ability to attract user engagement. Google achieved the highest average CTR (1.00%), substantially outperforming both TikTok (0.63%) and Facebook (0.57%).

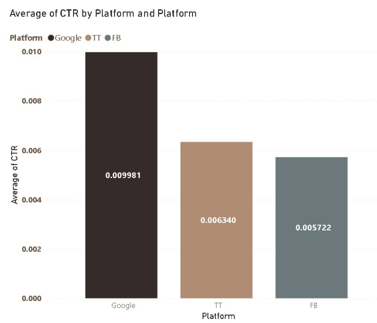


Figure 2.2: CTR by Platform

The statistical analysis confirmed that **these differences were significant** (Appendix B.1.2). These findings suggest that Google advertisements were more effective at capturing user attention and encouraging engagement.

2.2.3 Conversion Rate (CVR)

Conversion Rate (CVR) was used to measure each platform's effectiveness in converting user engagement into purchases. Facebook recorded the highest average CVR (2.87%), while Google and TikTok achieved nearly identical conversion rates (2.54%).

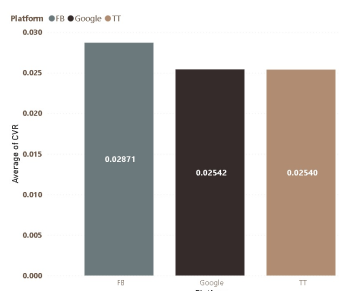


Figure 2.3: CVR by Platform

Statistical testing showed that **Facebook converted clicks into purchases more effectively than the other two platforms**, whereas **no statistically significant difference was observed between Google and TikTok** (Appendix B.1.3).

Although Facebook demonstrated stronger conversion efficiency, this metric should be interpreted alongside advertising cost and overall profitability.

2.2.4 Cost Per Click (CPC)

Cost Per Click (CPC) measures the average advertising cost required to generate a user click. Google achieved the lowest average CPC ($1.87), followed by TikTok ($1.98), while Facebook incurred the highest average CPC ($2.84).

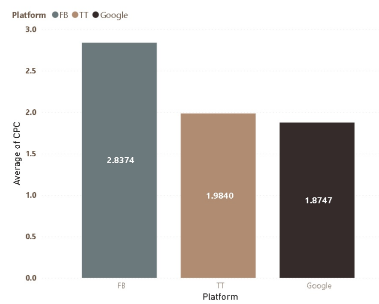


Figure 2.4: CPC by Platform

Statistical testing confirmed that the differences among all three platforms were significant (Appendix B.1.4). These results indicate that Google acquired user engagement more cost-effectively than either Facebook or TikTok.

2.2.5 Overall Platform Performance

Taken together, the four performance metrics indicate that **Google is currently the strongest-performing advertising platform** while Facebook is the weakest-performing advertising platform.

- Google generated the highest return on advertising investment(ROAS), attracted the greatest level of user engagement(CTR), and acquired clicks at the lowest cost(CPC).
- Although Facebook achieved the highest conversion rate, this advantage was offset by its substantially higher acquisition costs and lower overall ROAS.
- TikTok demonstrated balanced performance across all metrics but did not outperform Google in any of the primary business measures.
- From a business perspective, Google currently delivers the greatest overall value and represents the strongest candidate for additional advertising investment.
- Conversely, Facebook should not necessarily receive additional budget until improvements are made to reduce acquisition costs while maintaining its strong conversion performance.
- TikTok represents a stable intermediate option that may benefit from further optimization but currently offers lower overall returns than Google.

2.3 Regional Performance Analysis

The exploratory assessment suggested that advertising performance varied across geographic regions.

A complete summary of the statistical analyses, including assumption checks, Kruskal–Wallis tests, and Dunn's post-hoc comparisons, is provided in Appendix B.2.

2.3.1 Return on Advertising Spend (ROAS)

Return on Advertising Spend (ROAS) was used to evaluate advertising profitability across the four regions. The Northeast achieved the highest average ROAS (1.033), followed closely by the West (1.008), while the Midwest recorded the lowest average ROAS (0.834), as shown in *Fig. 2.5*.

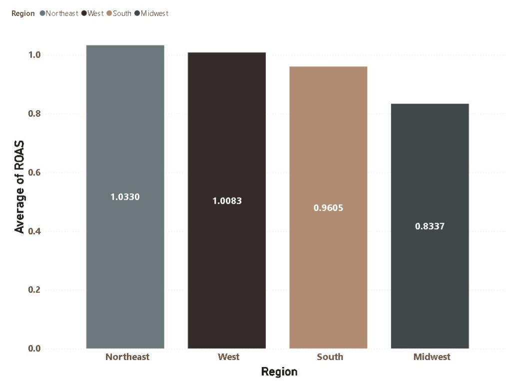


Figure 2.5: ROAS by Region

Statistical testing confirmed that the Midwest performed significantly worse than the other three regions (Appendix B.2.1). However, no statistically significant differences were observed among the Northeast, South, and West. These findings suggest that the Midwest consistently generated lower financial returns from advertising investment.

2.3.2 Click-Through Rate (CTR)

Click-Through Rate (CTR) was analyzed to assess audience engagement across regions. Although the Midwest recorded the highest average CTR (0.74%) and the Northeast the lowest (0.72%), these differences were minimal.

Figure 2.6: CTR by Region

Statistical testing found no significant differences in CTR among the four regions (Appendix B.2.2). This indicates that advertisements attracted similar levels of user engagement regardless of geographic region.

2.3.3 Conversion Rate (CVR)

Conversion Rate (CVR) was used to evaluate each region's ability to convert user engagement into purchases. The Northeast achieved the highest average CVR (2.84%), while the Midwest recorded the lowest average CVR (2.29%).

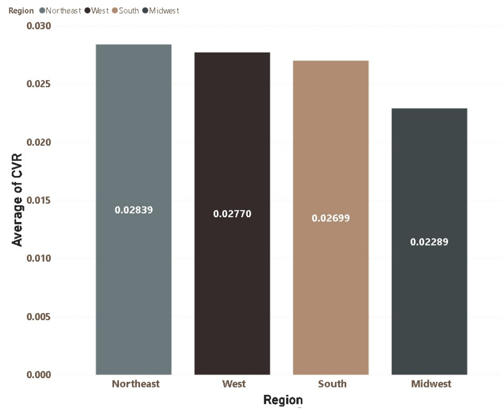


Figure 2.7: CVR by Region

Statistical testing confirmed that the Midwest converted significantly fewer clicks into purchases than every other region (Appendix B.2.3). The Northeast also achieved significantly higher conversion rates than both the South and West, whereas no statistically significant difference was observed between the South and West.

These findings indicate that regional differences are driven primarily by conversion effectiveness rather than customer engagement.

2.3.4 Cost Per Click (CPC)

Cost Per Click (CPC) measures the average advertising cost required to generate a user click. Average CPC values were nearly identical across all four regions, ranging from $2.24 to $2.25.

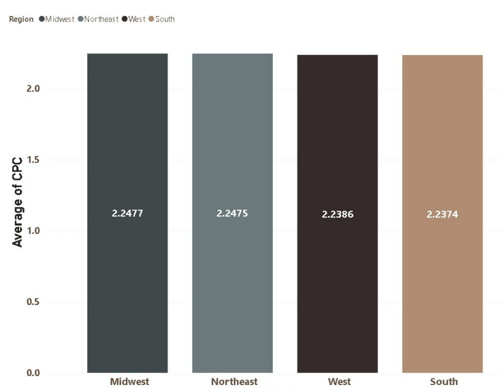


Figure 2.8: CPC by Region

Statistical testing confirmed that no significant regional differences existed in CPC (Appendix B.2.4). These results suggest that advertising costs were relatively consistent across geographic markets.

2.3.5 Regional Performance Summary

Taken together, the four performance metrics indicate that geographic location has a measurable impact on campaign profitability and conversion effectiveness, but relatively little influence on advertising engagement or acquisition cost.

- The **Northeast** demonstrated the strongest overall regional performance, achieving the highest ROAS and the highest conversion rate, suggesting that campaigns in this region generate the greatest financial return.
- The **Midwest** consistently underperformed, recording both the lowest ROAS and the lowest conversion rate despite exhibiting click-through rates comparable to the other regions. This suggests that the primary challenge lies in converting interested users into customers rather than attracting engagement.
- The **South** and **West** delivered comparable performance across all four metrics, with neither region showing a statistically significant advantage over the other in most comparisons.
- Because CTR and CPC did not differ significantly across regions, regional performance differences appear to be driven primarily by post-click behavior rather than differences in advertising cost or user engagement.

From a business perspective, further investigation should focus on **identifying factors that contribute to the Midwest's weaker conversion performance**.

Potential explanations include regional differences in customer preferences, purchasing behavior, market competition, or campaign targeting strategies.

2.4 Creative Performance Analysis

The exploratory assessment suggested that advertising performance varied across different creative types.

A complete summary of the statistical analyses, including assumption checks, Kruskal–Wallis tests, and Dunn's post-hoc comparisons, is provided in Appendix B.3.

2.4.1 Return on Advertising Spend (ROAS)

Return on Advertising Spend (ROAS) was used to evaluate the profitability of each creative type. Search advertisements achieved the highest average ROAS (1.324), followed by Video (0.989), while Image advertisements recorded the lowest average ROAS (0.618), as shown in *Fig.* **??**.

Statistical testing confirmed that Search advertisements significantly outperformed every other creative type (Appendix B.3.1). Carousel and Display creatives exhibited statistically similar ROAS values, while Display and Image creatives also showed no significant difference. These findings indicate that Search advertisements consistently generated superior financial returns compared with the remaining creative formats.

2.4.2 Click-Through Rate (CTR)

Click-Through Rate (CTR) was analyzed to evaluate audience engagement across creative types. Display and Search advertisements achieved the highest average CTR (approximately 1.00%), whereas Carousel advertisements recorded the lowest average CTR (0.54%), as illustrated in *Fig.* **??**.

images/ROAS_by_creative.jpg

Figure 2.9: ROAS by Creative Type

images/CTR_by_creative.jpg

Figure 2.10: CTR by Creative Type

Statistical testing indicated that Display and Search advertisements achieved statistically comparable click-through rates (Appendix B.3.2). Likewise, Carousel and Image creatives did not differ significantly from one another. All remaining pairwise comparisons were statistically significant, suggesting that Search and Display creatives consistently generated stronger audience engagement than the remaining formats.

2.4.3 Conversion Rate (CVR)

Conversion Rate (CVR) was used to evaluate the effectiveness of each creative type in converting user engagement into purchases. Carousel advertisements achieved the highest average conversion rate (3.00%), closely followed by Search advertisements (2.91%), whereas Display advertisements recorded the lowest average conversion rate (1.78%).

Statistical testing confirmed that Carousel and Search creatives achieved statistically equivalent conversion rates (Appendix B.3.3). Similarly, Image and Video creatives exhibited comparable images/CVR_by_creative.jpg

Figure 2.11: CVR by Creative Type

conversion performance, whereas Display advertisements converted significantly fewer clicks into purchases than every other creative type. These findings suggest that creative format has a substantial influence on post-click customer behavior.

2.4.4 Cost Per Click (CPC)

Cost Per Click (CPC) measures the average advertising cost required to generate a user click. Display advertisements achieved the lowest average CPC ($1.85), followed closely by Search advertisements ($1.88), while Carousel advertisements recorded the highest average CPC ($3.07).

images/CPC_by_creative.jpg

Figure 2.12: CPC by Creative Type

Statistical testing confirmed that Carousel and Image creatives exhibited statistically similar CPC values, while Display and Search advertisements also showed no statistically significant difference (Appendix B.3.4). All remaining pairwise comparisons were statistically significant, indicating meaningful differences in advertising cost among the creative formats.

2.4.5 Creative Performance Summary

Taken together, the four performance metrics indicate that creative type has a substantial influence on campaign profitability, audience engagement, conversion effectiveness, and advertising cost.

- **Search** advertisements demonstrated the strongest overall performance, achieving the highest ROAS while maintaining one of the lowest CPC values and a conversion rate comparable to the best-performing creative type. These findings suggest that Search creatives provide the most efficient balance between advertising cost and revenue generation.
- **Carousel** advertisements achieved the highest conversion rate but also incurred the highest CPC. This indicates that although Carousel creatives are effective at converting interested users, they require substantially greater advertising investment to generate those conversions.
- **Display** advertisements generated strong audience engagement through one of the highest CTR values and the lowest CPC. However, they consistently recorded the lowest conversion rate, suggesting that user interest generated by these advertisements does not effectively translate into purchases.
- **Image** advertisements exhibited the weakest overall financial performance, producing the lowest ROAS while incurring relatively high CPC values. This suggests that Image creatives provide comparatively poor returns on advertising investment.
- Because statistically significant differences were observed across all four performance metrics, creative selection appears to influence not only customer engagement but also advertising efficiency and overall campaign profitability.

From a business perspective, the results suggest that **Search advertisements should be prioritized when maximizing advertising profitability**, while Carousel creatives may be appropriate for campaigns focused on maximizing conversion rates despite their higher acquisition costs. Conversely, optimization efforts for Display advertisements should focus on improving post-click conversion performance, whereas Image creatives may benefit from redesign or replacement due to their comparatively weak financial performance.

Appendix A

Data Quality and Initial Exploratory Assessment

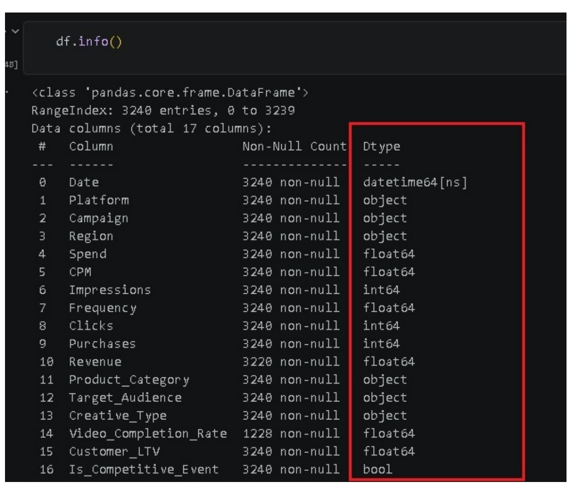
A.1 Data Quality Assessment

Prior to conducting business intelligence analysis, the advertising campaign dataset was evaluated to ensure its quality, reliability, and suitability for analysis. The objective of this assessment is to identify potential data quality issues, validate the performance metrics used throughout the project, examine unusual observations, and establish a set of baseline metrics that will guide subsequent business analysis.

The dataset contains daily advertising campaign performance across multiple platforms, regions, target audiences, product categories, and creative types during the first quarter of 2024.

A.1.1 Data Type

The data types of all variables were first examined to verify that each column was stored using an appropriate data type. As shown in *Fig. A.1*, numerical variables were stored as either `int64` or `float64`, categorical variables as `object`, the `Date` column as `datetime64[ns]`, and the `Is_Competitive_Event` variable as a boolean.



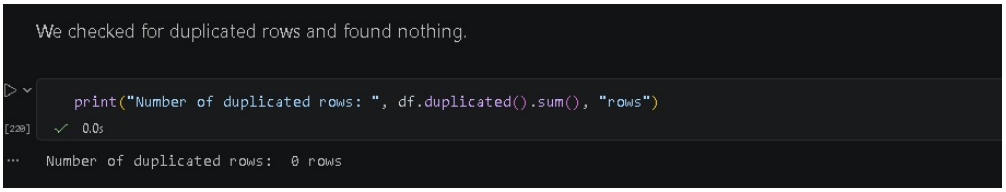
```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Date                   3240 non-null   datetime64[ns]
 1   Platform               3240 non-null   object
 2   Campaign               3240 non-null   object
 3   Region                 3240 non-null   object
 4   Spend                  3240 non-null   float64
 5   CPM                    3240 non-null   float64
 6   Impressions            3240 non-null   int64
 7   Frequency              3240 non-null   float64
 8   Clicks                 3240 non-null   int64
 9   Purchases              3240 non-null   int64
 10  Revenue                3220 non-null   float64
 11  Product_Category       3240 non-null   object
 12  Target_Audience        3240 non-null   object
 13  Creative_Type          3240 non-null   object
 14  Video_Completion_Rate  1228 non-null   float64
 15  Customer_LTV           3240 non-null   float64
 16  Is_Competitive_Event   3240 non-null   bool
```

Figure A.1: Checking datatypes of the dataframe

No inconsistencies in data types were identified; therefore, no type conversions were required.

A.1.2 Duplicate Records

The dataset was examined for duplicate observations.



We checked for duplicated rows and found nothing.

```
print("Number of duplicated rows: ", df.duplicated().sum(), "rows")
```

[220] 0.0s

```
Number of duplicated rows:  0 rows
```

Figure A.2: Checking for duplicate records

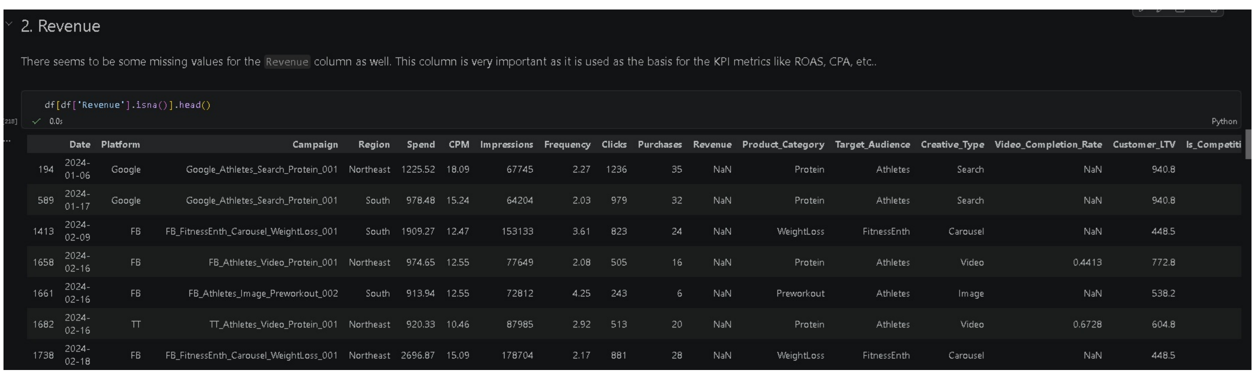
No duplicate records were identified as shown in *Fig. A.2*.

A.1.3 Data Completeness

The dataset was examined for missing values to determine whether incomplete observations could affect subsequent analyses.

- **Revenue**

  A sample of these datapoints are shown in *Fig. A.3*. It can be seen that twenty observations (approximately 0.6% of the dataset) contained missing revenue values.



2. Revenue

There seems to be some missing values for the Revenue column as well. This column is very important as it is used as the basis for the KPI metrics like ROAS, CPA, etc..

```
df[df['Revenue'].isna()].head()
```

[218] 0.0s Python

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competiti
194	2024-01-06	Google	Google_Athletes_Search_Protein_001	Northeast	1225.52	18.09	67745	2.27	1236	35	NaN	Protein	Athletes	Search	NaN	940.8	
589	2024-01-17	Google	Google_Athletes_Search_Protein_001	South	978.48	15.24	64204	2.03	979	32	NaN	Protein	Athletes	Search	NaN	940.8	
1413	2024-02-09	FB	FB_FitnessEnth_Carousel_WeightLoss_001	South	1909.27	12.47	153133	3.61	823	24	NaN	WeightLoss	FitnessEnth	Carousel	NaN	448.5	
1658	2024-02-16	FB	FB_Athletes_Video_Protein_001	Northeast	974.65	12.55	77649	2.08	505	16	NaN	Protein	Athletes	Video	0.4413	772.8	
1661	2024-02-16	FB	FB_Athletes_Image_Preworkout_002	South	913.94	12.55	72812	4.25	243	6	NaN	Preworkout	Athletes	Image	NaN	538.2	
1682	2024-02-16	TT	TT_Athletes_Video_Protein_001	Northeast	920.33	10.46	87985	2.92	513	20	NaN	Protein	Athletes	Video	0.6728	604.8	
1738	2024-02-18	FB	FB_FitnessEnth_Carousel_WeightLoss_001	Northeast	2696.87	15.09	178704	2.17	881	28	NaN	WeightLoss	FitnessEnth	Carousel	NaN	448.5	

Figure A.3: Sample datapoints with missing revenues.

  Because these represented a very small proportion of the data and no systematic pattern of missingness was identified, the corresponding observations were removed as shown in *Fig. A.4* prior to analysis.

- **Video Completion Rate**

  A large number of missing values were initially observed in the *Video Completion Rate* variable. Around 62% of the total dataset!

  Further investigation revealed that this metric is only applicable to **campaigns using video advertisements**. After restricting the dataframe to video creatives, **only 212 missing observations remained** as shown in *Fig. A.6*.

Because the proportion of missing values is very small, removing these observations is unlikely to introduce significant bias or substantially reduce the available information. Consequently, these rows will be removed from the dataset for analyses involving the `Revenue` variable.

```python
print("BEFORE DROPPING: Number of rows with missing revenue:",len(df[df['Revenue'].isna()]))

df.dropna(subset=["Revenue"], inplace=True)

print("AFTER DROPPING: Number of rows with missing revenue:",len(df[df['Revenue'].isna()]))
```

```
BEFORE DROPPING: Number of rows with missing revenue: 20
AFTER DROPPING: Number of rows with missing revenue: 0
```

Figure A.4: Dropped datapoints with missing revenues.

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Date                   3240 non-null   datetime64[ns]
 1   Platform               3240 non-null   object
 2   Campaign               3240 non-null   object
 3   Region                 3240 non-null   object
 4   Spend                  3240 non-null   float64
 5   CPM                    3240 non-null   float64
 6   Impressions            3240 non-null   int64
 7   Frequency              3240 non-null   float64
 8   Clicks                 3240 non-null   int64
 9   Purchases              3240 non-null   int64
 10  Revenue                3220 non-null   float64
 11  Product_Category       3240 non-null   object
 12  Target_Audience        3240 non-null   object
 13  Creative_Type          3240 non-null   object
 14  Video_Completion_Rate  1228 non-null   float64
 15  Customer_LTV           3240 non-null   float64
 16  Is_Competitive_Event   3240 non-null   bool
```

Figure A.5: `video_completion_rate` missing values

Consequently, these missing values were treated as **structurally missing** rather than data quality issues, and analyses involving this variable will use only the relevant subset of campaigns.

A.1.4 Logical Consistency Checks

Several validation checks were performed to identify logically inconsistent observations.

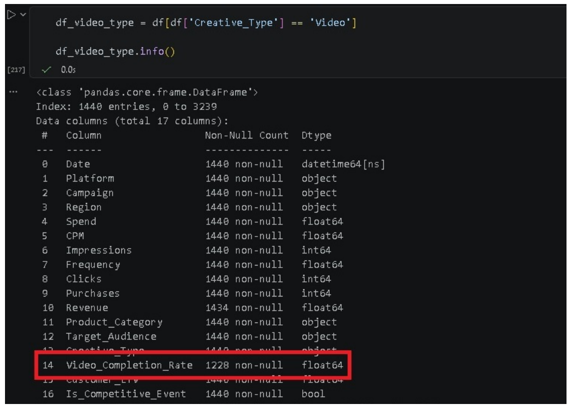
0 Clicks with non-zero purchase

I noticed that there were rows with 0 clicks but has non-zero Purchases, which I thought was weird at first. How are they able to purchase without clicking the ad.

Apparently, this happens in real world scenarios and they are called **view-through conversions**. People see the ad, they don't click it then later that evening, they remembered the brand and typed the website into google and bought the product.

But, at this point, I was wondering how they know which ad actually influenced the purchased without click or what if the buyer interacted with multiple ads before purchasing(cross-channel conversions). Apparently, this is related to **attribution**, and there are multiple attribution models like **last_click attribution**, **first-click attribution**, **linear_attribution**, etc.. which is not part of this data cleaning process.

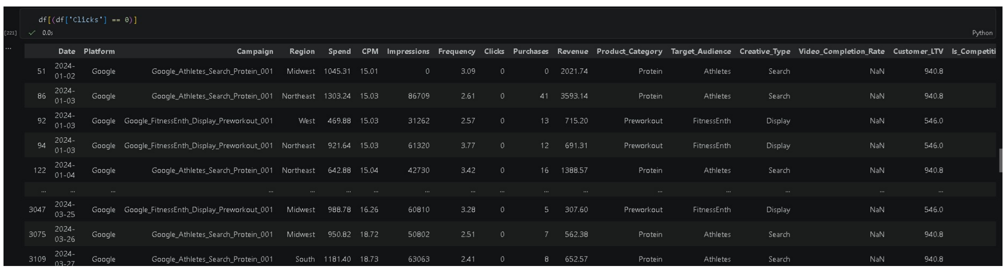
So, for now, I'll just assume that this dataset was collected using a certain attribution model and leave it at that. This justifies the 0 clicks with non-zero purchase.



```
df_video_type = df[df['Creative_Type'] == 'Video']

df_video_type.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1440 entries, 0 to 3239
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Date                   1440 non-null   datetime64[ns]
 1   Platform               1440 non-null   object
 2   Campaign               1440 non-null   object
 3   Region                 1440 non-null   object
 4   Spend                  1440 non-null   float64
 5   CPM                    1440 non-null   float64
 6   Impressions            1440 non-null   int64
 7   Frequency              1440 non-null   float64
 8   Clicks                 1440 non-null   int64
 9   Purchases              1440 non-null   int64
 10  Revenue                1434 non-null   float64
 11  Product_Category       1440 non-null   object
 12  Target_Audience        1440 non-null   object
 13  Creative_Type          1440 non-null   object
 14  Video_Completion_Rate  1228 non-null   float64
 15  Customer_LTV           1440 non-null   float64
 16  Is_Competitive_Event   1440 non-null   bool
```

Figure A.6: DataFrame filtered to video creative types



```
df[(df['Clicks'] == 0)]
```

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competiti
51	2024-01-02	Google	Google_Athletes_Search_Protein_001	Midwest	1045.31	15.01	0	3.09	0	0	2021.74	Protein	Athletes	Search	NaN	940.8	
86	2024-01-03	Google	Google_Athletes_Search_Protein_001	Northeast	1303.24	15.03	86709	2.61	0	41	3593.14	Protein	Athletes	Search	NaN	940.8	
92	2024-01-03	Google	Google_FitnessEnth_Display_Preworkout_001	West	469.88	15.03	31262	2.57	0	13	715.20	Preworkout	FitnessEnth	Display	NaN	546.0	
94	2024-01-03	Google	Google_FitnessEnth_Display_Preworkout_001	Northeast	921.64	15.03	61320	3.77	0	12	691.31	Preworkout	FitnessEnth	Display	NaN	546.0	
122	2024-01-04	Google	Google_Athletes_Search_Protein_001	Northeast	642.88	15.04	42730	3.42	0	16	1388.57	Protein	Athletes	Search	NaN	940.8	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
3047	2024-03-25	Google	Google_FitnessEnth_Display_Preworkout_001	Midwest	988.78	16.26	60810	3.28	0	5	307.60	Preworkout	FitnessEnth	Display	NaN	546.0	
3075	2024-03-26	Google	Google_Athletes_Search_Protein_001	Midwest	950.82	18.72	50802	2.51	0	7	562.38	Protein	Athletes	Search	NaN	940.8	
3109	2024-03-27	Google	Google_Athletes_Search_Protein_001	South	1181.40	18.73	63063	2.41	0	8	652.57	Protein	Athletes	Search	NaN	940.8	

Figure A.7: Sample datapoints with 0 clicks but non-zero purchase

> **Side Note:** After the data cleaning, all rows with zero clicks belong to the Google platform. (For the time being, this is just an observation. I'm noting it here so I don't forget to investigate it later.)

0 Impressions with non-zero Cost-Per-Mile(CPM)

Cost Per Mille (CPM) measures the advertising cost required to generate 1,000 impressions and is commonly used to evaluate the cost efficiency of awareness campaigns.

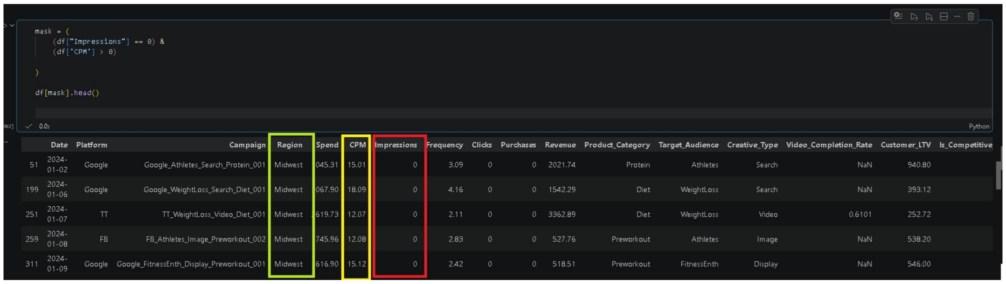
$$\text{CPM} = \frac{\text{Spend}}{\text{Impressions}} \times 1000 \tag{A.1}$$

Because impressions appear in the denominator, CPM is undefined when the number of impressions is zero.

During the data quality assessment, 47 observations (approximately 1.45% of the dataset) were identified with `Impressions = 0` but non-zero with CPM values.

Furthermore, all of these observations originated from the Midwest region as shown in *Fig. A.8*, suggesting a systematic reporting inconsistency rather than isolated data entry errors.

Since these observations violate the definition of CPM and the 0 impressions value could affect the calculation of other performance metrics, such as CTR and CVR, they were removed from the dataset prior to analysis.



```python
mask = (
    (df["Impressions"] == 0) &
    (df['CPM'] > 0)

)

df[mask].head()
```

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competitive
51	2024-01-02	Google	Google_Athletes_Search_Protein_001	Midwest	045.31	15.01	0	3.09	0	0	2021.74	Protein	Athletes	Search	NaN	940.80	
199	2024-01-06	Google	Google_WeightLoss_Search_Diet_001	Midwest	067.90	18.09	0	4.16	0	0	1542.29	Diet	WeightLoss	Search	NaN	393.12	
251	2024-01-07	TT	TT_WeightLoss_Video_Diet_001	Midwest	619.73	12.07	0	2.11	0	0	3362.89	Diet	WeightLoss	Video	0.6101	252.72	
259	2024-01-08	FB	FB_Athletes_Image_Preworkout_002	Midwest	745.96	12.08	0	2.83	0	0	527.76	Preworkout	Athletes	Image	NaN	538.20	
311	2024-01-09	Google	Google_FitnessEnth_Display_Preworkout_001	Midwest	616.90	15.12	0	2.42	0	0	518.51	Preworkout	FitnessEnth	Display	NaN	546.00	

Figure A.8: Sample rows that has 0 impressions with non-zero CPM

```python
mask = (df["Impressions"] == 0) & (df["CPM"] > 0)

print("Rows before removal:", mask.sum())

df.drop(df[mask].index, inplace=True)

mask = (df["Impressions"] == 0) & (df["CPM"] > 0)
print("Rows after removal:", mask.sum())
```

```
Rows before removal: 47
Rows after removal: 0
```

Figure A.9: Dropped Rows that has 0 impressions with non-zero CPM

A.1.5 Outlier Assessment

Boxplots were used to identify potential outliers across the numerical variables as shown in *Fig. A.10*. Severe outliers were identified for the Spend column and will be investigated in this section.

`Spend` severe outliers investigation

First, I listed the 5 datapoints with severe `Spend` outliers as shown in *Fig. A.12*. I noticed that in this 5 datapoints, the **CPM do not match the Ad Spend and Impression**. This tells me that there is a possibility that there are more rows with mismatched CPM.

So, I checked and found out that only these 5 rows have mismatched CPM as shown in *Fig. A.11*, after dropping the rows from earlier(the ones with 0 impression,clicks and purchases).

Then, I plotted the `Spend` boxplots of these Campaign Ads.

From this analysis, I noticed the following consistencies:

1. The reported CPM values(from datapoints with severe `Spend` outliers) were exactly one-hundredth of the values calculated from the corresponding Spend and Impressions.

2. The calculated CPM values do not match the 'Impressions' and 'Spend' columns, and this is limited only to the 5 reported rows with severe 'Spend' outliers.

3. Inspection of the affected campaigns showed that the unusually large Spend values occurred only on a single day and were not associated with competitive events.

Since the discrepancy was systematic (a factor of 100) and limited to these five observations, the Spend values were divided by 100 to restore consistency with the reported CPM values.

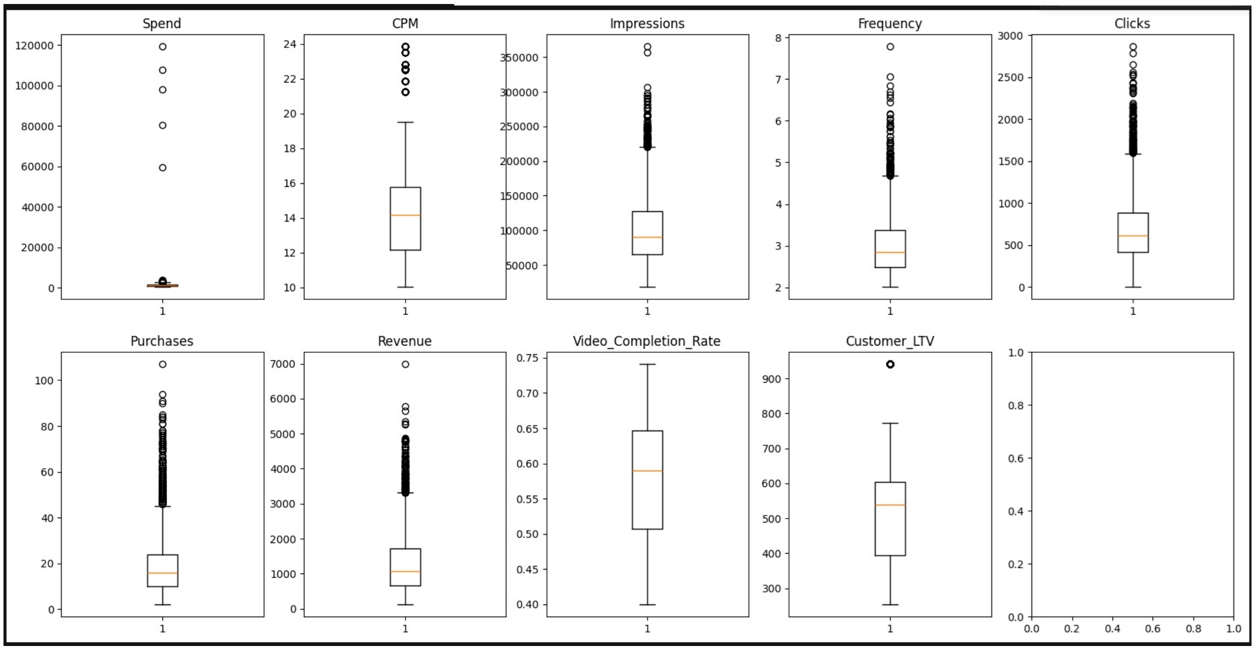


Figure A.10: Boxplots for all numerical columns of the dataframe

A.1.6 Data Cleaning Summary

The data cleaning process consisted of:

- Verified variable data types
- Verifying duplicate observations
- Removing observations(20 datapoints) with missing revenue.
- Retaining structurally missing Video Completion Rate values.
- Retaining valid view-through conversions.
- Removing observations(47 datapoints) with 0 clicks,impressions and purchases with non-zero purchase.
- Addressed severe `Spend` outliers.

Overall, the dataset was determined to be sufficiently complete and reliable for subsequent analysis.

```python
import numpy as np

df["Calculated_CPM"] = (df["Spend"] / df["Impressions"]) * 1000

# Compare with a tolerance of 0.01
matches = np.isclose(df["Calculated_CPM"], df["CPM"], atol=0.1)

mismatches = df[~matches]

print(f"Number of CPM mismatches: {len(mismatches)}")

mismatches[["Campaign","Spend", "Impressions", "CPM", "Calculated_CPM"]].sort_values(by = 'Spend', ascending=False)
```

[312] 0.0s

```
Number of CPM mismatches: 5
```

	Campaign	Spend	Impressions	CPM	Calculated_CPM
1286	TT_Athletes_Video_Protein_001	119315.0	115279	10.35	1035.010713
1425	Google_FitnessEnth_Display_Preworkout_001	107742.0	69131	15.58	1558.519333
2892	Google_Athletes_Search_Protein_001	97883.0	60421	16.20	1620.016220
2744	FB_FitnessEnth_Carousel_WeightLoss_001	80592.0	52013	15.49	1549.458789
923	Google_FitnessEnth_Display_Preworkout_001	59556.0	33683	17.68	1768.132292

Figure A.11: Number of rows with CPM mismatch

```python
df.loc[outlier_masks["Spend"], "Spend"] /=100

df[outlier_masks["Spend"]].sort_values(by = 'Spend', ascending=False).head(5)
```

0.0s

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate
1286	2024-02-05	TT	TT_Athletes_Video_Protein_001	Northeast	1193.15	10.35	115279	2.97	705	21	1746.21	Protein	Athletes	Video	0.6891
1425	2024-02-09	Google	Google_FitnessEnth_Display_Preworkout_001	South	1077.42	15.58	69131	2.99	798	11	602.20	Preworkout	FitnessEnth	Display	NaN
2892	2024-03-21	Google	Google_Athletes_Search_Protein_001	West	978.83	16.20	60421	2.35	399	16	1325.27	Protein	Athletes	Search	NaN
2744	2024-03-17	FB	FB_FitnessEnth_Carousel_WeightLoss_001	West	805.92	15.49	52013	2.36	145	2	125.28	WeightLoss	FitnessEnth	Carousel	NaN
923	2024-01-26	Google	Google_FitnessEnth_Display_Preworkout_001	Midwest	595.56	17.68	33683	3.44	517	7	414.71	Preworkout	FitnessEnth	Display	NaN

Figure A.12: Datapoints with severe `Spend` outliers

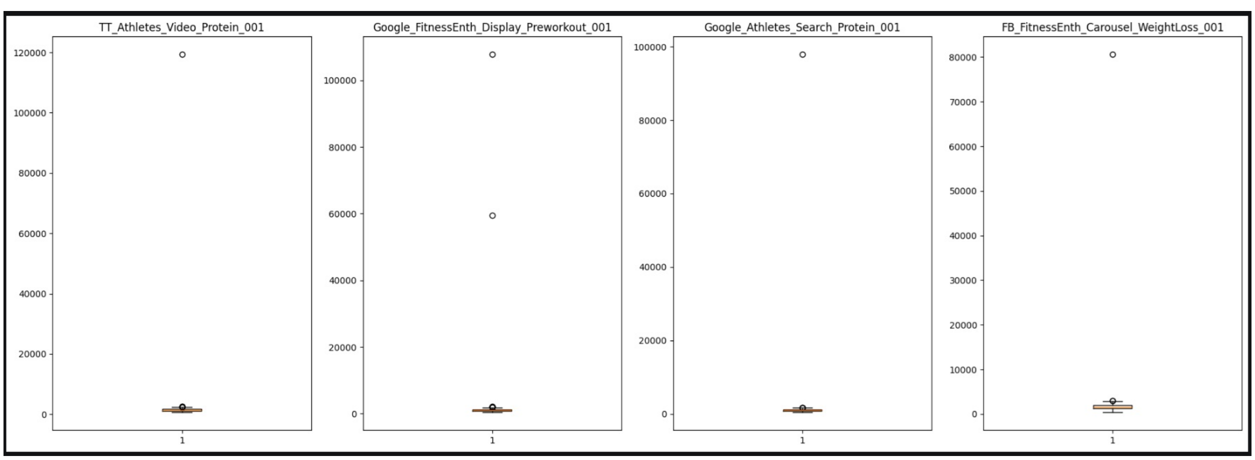


Figure A.13: `Spend` boxplots for campaign ads with severe `Spend` outliers

Appendix B

Statistical Test Results

B.1 Channel performance analysis

B.1.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.1: Descriptive statistics for ROAS by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	0.8162	0.7499	0.4225	1054
Google	1.1057	0.9456	0.6495	1053
TikTok	0.9602	0.8625	0.4950	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **ROAS differed significantly** across platforms.

Kruskal–Wallis Test

Statistic	Value
H-statistic	106.6989
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in ROAS across advertising platforms.

Dunn Post-hoc Test

Table B.2: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	0.0002
TikTok	< 0.0001	0.0002	–

Summary

- The highest average ROAS was achieved by Google (1.1057).
- The lowest average ROAS was observed for FB (0.8162).
- Differences of ROAS between platforms are statistically significant ($p < 0.05$).

B.1.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.3: Descriptive statistics for CTR by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	0.0057	0.0056	0.0021	1054
Google	0.0100	0.0098	0.0052	1053
TikTok	0.0063	0.0062	0.0018	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of CTR **differed significantly across platforms**.

Kruskal–Wallis Test

Statistic	Value
H-statistic	634.6820
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CTR across advertising platforms.

Dunn Post-hoc Test

Table B.4: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	< 0.0001
TikTok	< 0.0001	< 0.0001	–

Summary

- The highest average CTR was achieved by Google (0.0100).
- The lowest average CTR was observed for FB (0.0057).
- Differences of CTR between platforms are statistically significant ($p < 0.05$).

B.1.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.5: Descriptive statistics for CVR by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	0.0287	0.0277	0.0076	1054
Google	0.0254	0.0254	0.0082	954
TikTok	0.0254	0.0258	0.0081	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions **significantly deviated from normality** ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	0.0072

The variability of CVR **differed significantly across platforms**.

Kruskal–Wallis Test

Statistic	Value
H-statistic	96.7108
p-value	< 0.0001

The Kruskal–Wallis test indicates **that at least one platform differs significantly** in CVR.

Dunn Post-hoc Test

Table B.6: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	1.0000
TikTok	< 0.0001	1.0000	–

Facebook differed significantly from both Google and TikTok, whereas **no statistically significant difference** was observed between Google and TikTok.

Summary

- The **highest average CVR** was achieved by **Facebook** (0.0287).
- The **lowest average CVR** was observed for **Google** (0.0254), although Google and TikTok had nearly identical average CVRs.
- **Facebook differed significantly** from both Google and TikTok, while **no statistically significant difference** was observed between Google and TikTok.

B.1.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.7: Descriptive statistics for CPC by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	2.8374	2.5049	1.3116	1054
Google	1.8747	1.5936	0.9076	954
TikTok	1.9840	1.8452	0.6718	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CPC differed significantly** across platforms.

Kruskal–Wallis Test

Statistic	Value
H-statistic	447.1995
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CPC across advertising platforms.

Dunn Post-hoc Test

Table B.8: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	< 0.0001
TikTok	< 0.0001	< 0.0001	–

Summary

- The highest average CPC was achieved by Facebook (2.8374).
- The lowest average CPC was observed for Google (1.8747).
- Differences of CPC between platforms are statistically significant ($p < 0.05$).

B.2 Regional performance comparison

B.2.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.9: Descriptive statistics for ROAS by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.8337	0.6951	0.4940	760
Northeast	1.0330	0.8678	0.5960	803
South	0.9605	0.8535	0.4868	803
West	1.0083	0.8915	0.5662	807

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of ROAS differed significantly across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	69.2321
p-value	< 0.0001

The Kruskal–Wallis test indicates that at least one region differs significantly in ROAS.

Dunn Post-hoc Test

Table B.10: Pairwise Dunn test p-values (Bonferroni adjusted).

	Midwest	Northeast	South	West
Midwest	–	< 0.0001	< 0.0001	< 0.0001
Northeast	< 0.0001	–	1.0000	1.0000
South	< 0.0001	1.0000	–	1.0000
West	< 0.0001	1.0000	1.0000	–

The Midwest **differed significantly** from the Northeast, South, and West, whereas **no statistically significant differences** were observed among the Northeast, South, and West.

Summary

- The highest average ROAS was achieved by the Northeast (1.0330).
- The lowest average ROAS was observed for the Midwest (0.8337).
- The Midwest differed significantly from all other regions, while no statistically significant differences were observed among the Northeast, South, and West.

B.2.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.11: Descriptive statistics for CTR by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.0074	0.0066	0.0039	760
Northeast	0.0072	0.0064	0.0039	803
South	0.0074	0.0067	0.0038	803
West	0.0073	0.0067	0.0038	807

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions **significantly deviated** from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	0.8844

The variability of CTR **did not differ significantly** across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	0.9848
p-value	0.8049

The Kruskal–Wallis test indicates **no statistically significant differences** in CTR across regions.

Summary

- The highest average CTR was achieved by the Midwest (0.0074), although the regional averages were very similar.
- The lowest average CTR was observed for the Northeast (0.0072).
- No statistically significant differences in CTR were found between regions ($p \geq 0.05$).

B.2.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.12: Descriptive statistics for CVR by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.0229	0.0236	0.0073	737
Northeast	0.0284	0.0288	0.0086	771
South	0.0270	0.0270	0.0071	785
West	0.0277	0.0267	0.0081	781

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions **significantly deviated** from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of CVR **differed significantly** across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	190.4208
p-value	< 0.0001

The Kruskal–Wallis test indicates that at least one region **differs significantly** in CVR.

Dunn Post-hoc Test

Table B.13: Pairwise Dunn test p-values (Bonferroni adjusted).

	Midwest	Northeast	South	West
Midwest	–	< 0.0001	< 0.0001	< 0.0001
Northeast	< 0.0001	–	0.0057	0.0396
South	< 0.0001	0.0057	–	1.0000
West	< 0.0001	0.0396	1.0000	–

The Midwest **differed significantly** from all other regions. The Northeast also differed significantly from both the South and the West, whereas **no statistically significant difference** was observed between the South and the West.

Summary

- The highest average CVR was achieved by the Northeast (0.0284).
- The lowest average CVR was observed for the Midwest (0.0229).
- The Midwest differed significantly from all other regions, while the South and West did not differ significantly from one another.

B.2.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.14: Descriptive statistics for CPC by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	2.2477	1.9809	1.1166	737
Northeast	2.2475	1.9919	1.0984	771
South	2.2374	2.0091	1.0791	785
West	2.2386	1.9872	1.0686	781

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions **significantly deviated** from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	0.9460

The variability of CPC did **not differ significantly** across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	0.0346
p-value	0.9983

The Kruskal–Wallis test indicates **no statistically significant differences** in CPC across regions.

Summary

- The highest average CPC was observed in the Midwest (2.2477), although the regional averages were nearly identical.
- The lowest average CPC was observed in the South (2.2374).
- No statistically significant differences in CPC were found between regions ($p \geq 0.05$).

B.3 Creative performance analysis

B.3.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.15: Descriptive statistics for ROAS by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	0.7560	0.6572	0.4480	355
Display	0.6626	0.6194	0.2685	348
Image	0.6176	0.5656	0.2749	349
Search	1.3244	1.2455	0.6708	705
Video	0.9886	0.9058	0.4730	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	< 0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **ROAS differed significantly** across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	606.8404
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in ROAS across creative types.

Dunn Post-hoc Test

Table B.16: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	0.2527	0.0008	< 0.0001	< 0.0001
Display	0.2527	–	0.8940	< 0.0001	< 0.0001
Image	0.0008	0.8940	–	< 0.0001	< 0.0001
Search	< 0.0001	< 0.0001	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Summary

- The highest average ROAS was achieved by **Search** (1.3244).
- The lowest average ROAS was observed for **Image** (0.6176).
- **Search differed significantly** from all other creative types. No statistically significant differences were observed between **Carousel and Display**, or between **Display and Image**.

B.3.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.17: Descriptive statistics for CTR by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	0.0054	0.0051	0.0021	355
Display	0.0100	0.0098	0.0054	348
Image	0.0055	0.0052	0.0022	349
Search	0.0100	0.0099	0.0050	705
Video	0.0063	0.0061	0.0018	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	< 0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CTR differed significantly** across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	658.6765
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CTR across creative types.

Dunn Post-hoc Test

Table B.18: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	< 0.0001	1.0000	< 0.0001	< 0.0001
Display	< 0.0001	–	< 0.0001	1.0000	< 0.0001
Image	1.0000	< 0.0001	–	< 0.0001	< 0.0001
Search	< 0.0001	1.0000	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Summary

- The highest average CTR was achieved by **Display** (0.0100), closely followed by **Search** (0.0100).
- The lowest average CTR was observed for **Carousel** (0.0054).
- Display and Search did not differ significantly, while Carousel and Image also showed no statistically significant difference. All remaining pairwise comparisons were statistically significant.

B.3.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.19: Descriptive statistics for CVR by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	0.0300	0.0289	0.0078	355
Display	0.0178	0.0169	0.0054	310
Image	0.0265	0.0252	0.0081	349
Search	0.0291	0.0284	0.0066	644
Video	0.0264	0.0266	0.0079	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CVR differed significantly** across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	489.5054
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CVR across creative types.

Dunn Post-hoc Test

Table B.20: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	< 0.0001	< 0.0001	1.0000	< 0.0001
Display	< 0.0001	–	< 0.0001	< 0.0001	< 0.0001
Image	< 0.0001	< 0.0001	–	< 0.0001	1.0000
Search	1.0000	< 0.0001	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	1.0000	< 0.0001	–

Carousel and Search did not differ significantly, while Image and Video also showed no statistically significant difference. All remaining pairwise comparisons were statistically significant.

Summary

- The **highest average CVR** was achieved by **Carousel** (0.0300), closely followed by **Search** (0.0291).
- The **lowest average CVR** was observed for **Display** (0.0178).
- Carousel and Search exhibited statistically similar CVRs, while Image and Video also did not differ significantly. All other pairwise comparisons showed statistically significant differences.

B.3.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.21: Descriptive statistics for CPC by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	3.0715	2.6470	1.4731	355
Display	1.8540	1.5707	0.9053	310
Image	3.0082	2.6030	1.4140	349
Search	1.8847	1.6315	0.9092	644
Video	2.0942	1.9452	0.7485	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	< 0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CPC differed significantly** across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	409.9110
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CPC across creative types.

Dunn Post-hoc Test

Table B.22: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	< 0.0001	1.0000	< 0.0001	< 0.0001
Display	< 0.0001	–	< 0.0001	1.0000	< 0.0001
Image	1.0000	< 0.0001	–	< 0.0001	< 0.0001
Search	< 0.0001	1.0000	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Carousel and Image did not differ significantly, while Display and Search also exhibited statistically similar CPC values. All remaining pairwise comparisons were statistically significant.

Summary

- The **highest average CPC** was achieved by **Carousel** (3.0715), followed closely by **Image** (3.0082).
- The **lowest average CPC** was observed for **Display** (1.8540).
- Carousel and Image exhibited statistically similar CPC values, while Display and Search also did not differ significantly. All other pairwise comparisons showed statistically significant differences.